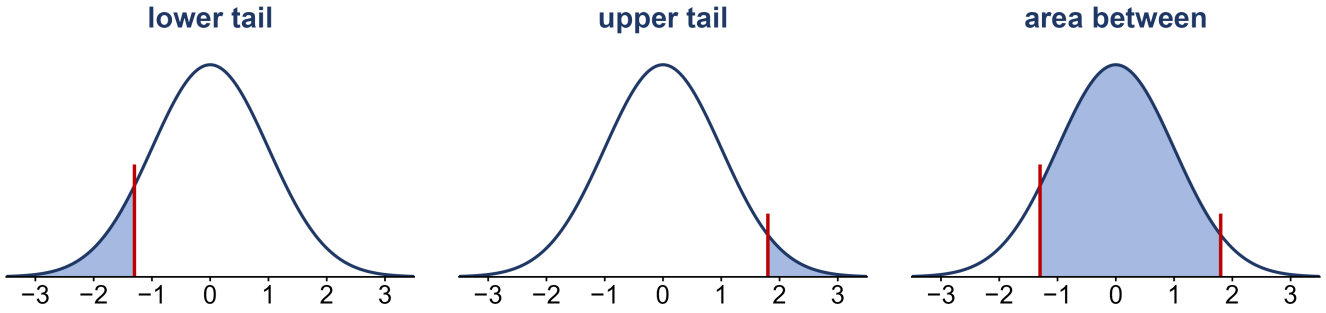
Week 9 Notes

STAT 251 SECTION 01

Finding normal probabilities

In Week 7 we said that for a continuous random variable, probability is **area under the density curve**, and that a single value has probability zero, $P(X = c) = 0$. So the questions we ask about a continuous variable are always about an *interval*: below some value, above it, or between two values. For the normal distribution those areas come from the **standard normal table** (the z-table), which we first used in Week 6.

The catch is that the table gives only **one** kind of area: the area to the **left** of z, $P(Z \le z)$, where $Z \sim N(0, 1)$ is the standard normal. Every other area has to be built out of that one.



- **Lower tail.** $P(Z < z)$ is the table value itself.
- **Upper tail.** The whole area under the curve is 1, so the area to the right is what is left over:

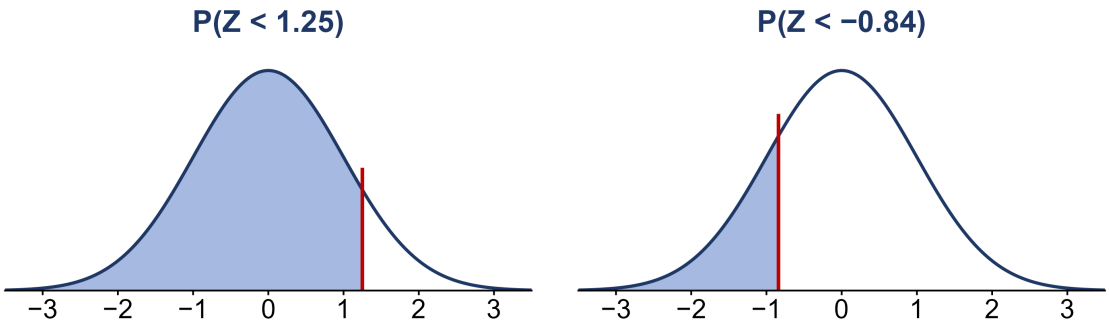
$$P(Z > z) = 1 - P(Z \le z)$$

- **Area between.** Take everything to the left of the larger value, and remove everything to the left of the smaller one:

$$P(a < Z < b) = P(Z \le b) - P(Z \le a)$$

Because $P(Z = z) = 0$, it does not matter whether the inequality is $<$ or $\le$. Always **sketch the curve and shade the area first**: the picture tells you which rule to use, and whether your answer should be big or small.

The lower tail

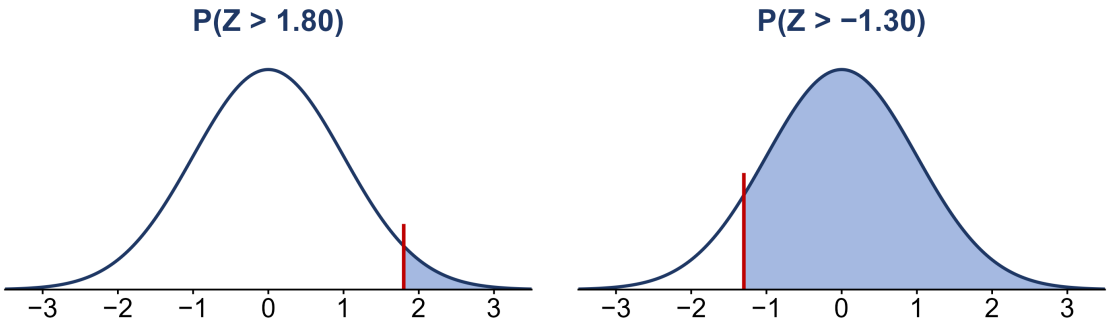


Example: lower tail

a. $P(Z < 1.25)$. Shade everything to the left of 1.25. Since 1.25 is above the mean of 0, the shaded area is most of the curve, so expect an answer well over one half. This is a lower tail, so the table value is the answer. Split 1.25 into $1.2 + 0.05$: row 1.2, column .05 gives $P(Z < 1.25) = \mathbf{0.8944}$.

b. $P(Z < -0.84)$. Shade everything to the left of -0.84. That is a small slice of the left side, less than one half, because a negative z is below the mean. It is again a lower tail: row -0.8, column .04 gives $P(Z < -0.84) = \mathbf{0.2005}$.

The upper tail



Example: upper tail

a. $P(Z > 1.80)$. Shade everything to the right of 1.80, a thin tail, so expect a small answer. The table gives the area to the **left**, so look that up first: row 1.8, column .00 gives $P(Z \leq 1.80) = 0.9641$. The whole area is 1, so

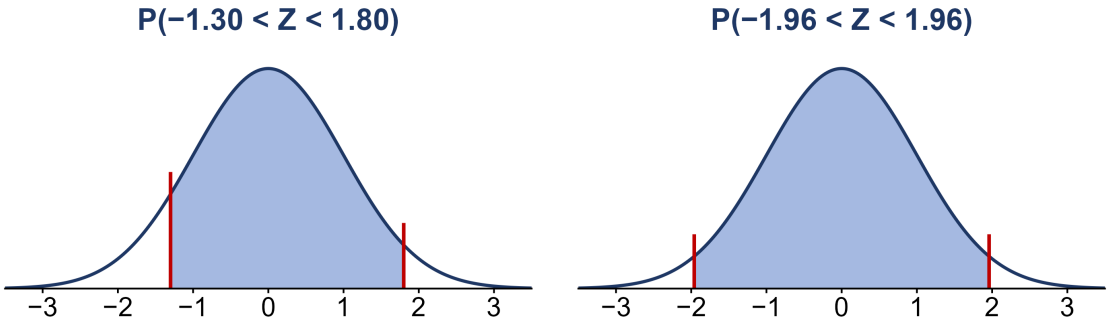
$$P(Z > 1.80) = 1 - P(Z \leq 1.80) = 1 - 0.9641 = \mathbf{0.0359}$$

b. $P(Z > -1.30)$. Shade everything to the right of -1.30, which is most of the curve. Row -1.3, column .00 gives $P(Z \leq -1.30) = 0.0968$, so

$$P(Z > -1.30) = 1 - 0.0968 = \mathbf{0.9032}$$

The normal curve is symmetric, so the area to the right of -1.30 is the same as the area to the left of $+1.30$: the table gives $P(Z \leq 1.30) = 0.9032$, the same answer.

The area between



Example: area between

a. $P(-1.30 < Z < 1.80)$. Shade the area between the two values. Take everything to the left of the larger value (row 1.8, column .00: 0.9641), and remove everything to the left of the smaller value (row -1.3, column .00: 0.0968):

$$P(-1.30 < Z < 1.80) = 0.9641 - 0.0968 = \mathbf{0.8673}$$

b. $P(-1.96 < Z < 1.96)$. Row 1.9, column .06 gives 0.9750, and row -1.9, column .06 gives 0.0250:

$$P(-1.96 < Z < 1.96) = 0.9750 - 0.0250 = \mathbf{0.9500}$$

Remember that second one. Exactly 95% of a normal distribution lies within **1.96** standard deviations of the mean; the empirical rule's "2" is that number rounded. It is where the 2 in a 95% confidence interval comes from.

Practice

For $Z \sim N(0, 1)$, sketch and shade each area, then use the table:

a. $P(Z \leq 2.00)$ (b) $P(Z > 0.50)$ (c) $P(-1 < Z < 1)$ (d) $P(-2 < Z < 2)$

Worked answers:

- a. A **lower tail**, so the table value is the answer: row 2.0, column .00 gives $P(Z \leq 2.00) = 0.9772$.

- b. An **upper tail**: row 0.5, column .00 gives $P(Z \leq 0.50) = 0.6915$, so $P(Z > 0.50) = 1 - 0.6915 = 0.3085$.
- c. An **area between**: row 1.0 gives 0.8413 and row -1.0 gives 0.1587, so $P(-1 < Z < 1) = 0.8413 - 0.1587 = 0.6826$, about 68%.
- d. An **area between**: row 2.0 gives 0.9772 and row -2.0 gives 0.0228, so $P(-2 < Z < 2) = 0.9772 - 0.0228 = 0.9544$, about 95%.

c. and (d) are the empirical rule from Week 6: about 68% of a normal distribution lies within one standard deviation of the mean and about 95% within two.

Any normal distribution

The same three rules work for any normal variable $X \sim N(\mu, \sigma)$: first convert each value to a z-score, $z = (x - \mu)/\sigma$, then find the area for Z. For example, if $X \sim N(5, 10)$, then $P(X \geq 25) = P(Z \geq 2) = 1 - 0.9772 = 0.0228$. The central limit theorem, next, uses exactly this, with the standard error in place of σ.

The central limit theorem and an interval for a mean

This section works through every example from the Lecture 19 slides. The longer write-up, with the simulations behind the theorem, is on the Week 8 notes page, starting at "The central limit theorem" (lecturemenu8.html#the-central-limit-theorem).

The theorem

In Week 8 we found the **center** and **spread** of the sampling distribution of $\bar{x}$: mean μ and standard error $\sigma/\sqrt{n}$. The **central limit theorem** (CLT) gives its **shape**:

- **Central limit theorem.** Take random samples of size n from **any** population with mean μ and standard deviation σ. If n is large enough, the sampling distribution of $\bar{x}$ is approximately normal:

$$\bar{x} \sim N\left(\mu,\ \frac{\sigma}{\sqrt{n}}\right)$$

A sample proportion is a sample mean of 0s and 1s, so the same is true of $\hat{p}$:

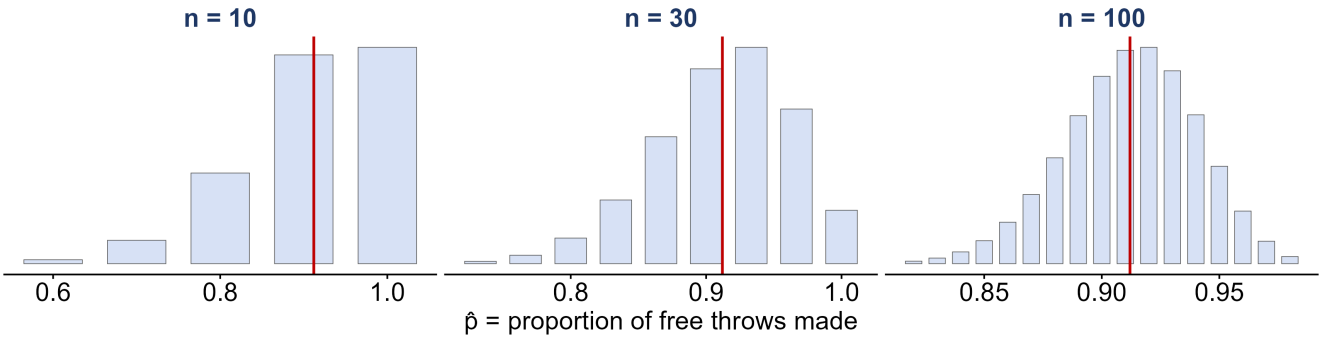
$$\hat{p} \sim N\left(p,\ \sqrt{\frac{p(1-p)}{n}}\right)$$

The surprising part is "**any** population". Even when the population is skewed, like the minutes until the next goal at the World Cup, averages of enough observations are bell shaped. The larger n is, the better the approximation. The textbook's guides for "large enough" are:

- $n \geq 30$ for $\bar{x}$. A population that is close to normal needs less; one that is very skewed or has extreme outliers may need more.
- $np > 5$ and $n(1-p) > 5$ for $\hat{p}$, so there are enough successes *and* failures.

Example: the CLT for a proportion

Stephen Curry has made 4,236 of his 4,645 career free throws, so take $p = 0.912$. Let $\hat{p}$ be the proportion he makes over a stretch of n free throws. Below are the exact sampling distributions of $\hat{p}$ for $n = 10$, 30 and 100. For which n is $n(1-p) > 5$, and which distributions look like a bell?



Curry almost always makes his free throws, so the check that can fail is the one for **misses**, $n(1-p)$, with $1 - p = 0.088$:

- $n = 10$: $10 \times 0.088 = 0.88$, which fails. The distribution is piled up against 1 and strongly skewed.
- $n = 30$: $30 \times 0.088 = 2.64$, which still fails. It is still skewed left.
- $n = 100$: $100 \times 0.088 = 8.8 > 5$, which passes. This is the only one close to a bell.

The check and the pictures agree. The CLT needs enough successes *and* enough failures; with too few expected misses, $\hat{p}$ cannot spread out to the right of p and the distribution stays lopsided.

Using it

Because $\bar{x}$ is approximately normal, we can find probabilities about it with the z-table, exactly as in the first part of these notes. The one change is that we standardize with the **standard error**, not σ:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

In particular, about 95% of the time a statistic lands within **2 standard errors** of its parameter:

$$P\left(\mu - 2\frac{\sigma}{\sqrt{n}} \;\le\; \bar{x} \;\le\; \mu + 2\frac{\sigma}{\sqrt{n}}\right) \approx 0.95$$

Example: free throws, again

Over a stretch of $n = 100$ free throws, let $\hat{p}$ be the proportion Curry makes, with $p = 0.912$. We just checked that $n(1-p) = 8.8 > 5$ (and $np = 91.2 > 5$), so by the CLT $\hat{p}$ is approximately normal.

a. *What are the mean and standard error of $\hat{p}$?* The mean of $\hat{p}$ is p itself, **0.912**. The standard error is

$$\sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.912 \times 0.088}{100}} = \sqrt{0.000803} \approx \mathbf{0.028}$$

b. *About 95% of the time, $\hat{p}$ falls between what two values?* Within 2 standard errors of its mean:

$$0.912 \pm 2(0.028) = 0.912 \pm 0.056, \quad \text{from } \mathbf{0.856} \text{ to } \mathbf{0.968}$$

So in about 95% of stretches of 100 free throws, Curry makes between about 86 and 97 of them.

Example: a tournament's average

The number of goals in a World Cup match is $X \sim \text{Poisson}(\lambda = 2.69)$, the 2022 rate. A tournament has 64 matches, so the tournament's average number of goals per match is a sample mean with $n = 64$. For a Poisson variable the mean and the variance are both λ, so

$$\mu = 2.69, \qquad \sigma = \sqrt{2.69} \approx 1.64$$

a. *What is the standard error of a tournament's average?*

$$\frac{\sigma}{\sqrt{n}} = \frac{1.64}{\sqrt{64}} = \frac{1.64}{8} = \mathbf{0.205}$$

b. *About 95% of tournaments average between what two values?* $n = 64 \geq 30$, so the CLT says $\bar{x}$ is approximately normal, and about 95% of tournaments land within 2 standard errors of μ:

$$2.69 \pm 2(0.205) = 2.69 \pm 0.41, \quad \text{from } \mathbf{2.28} \text{ to } \mathbf{3.10} \text{ goals per match}$$

c. *How likely is a tournament that averages 3 or more goals per match?* Sketch the normal curve for $\bar{x}$ centered at 2.69 and shade the area to the right of 3, a small upper tail. Standardize with the standard error:

$$z = \frac{3 - 2.69}{0.205} = \frac{0.31}{0.205} \approx 1.51$$

Row 1.5, column .01 of the table gives $P(Z \leq 1.51) = 0.9345$, so

$$P(\bar{x} \geq 3) = P(Z \geq 1.51) = 1 - 0.9345 = \mathbf{0.0655}$$

About a 7% chance. A single match can easily have 0 goals or 6; a tournament's average is far more predictable.

A 95% confidence interval for a mean

If $\bar{x}$ is within 2 standard errors of μ, then μ is within 2 standard errors of $\bar{x}$. Turning the statement around gives the **95% confidence interval** for μ:

$$\bar{x} \ \pm \ 2\frac{\sigma}{\sqrt{n}}$$

which has the form **point estimate $\pm$ margin of error**.

The data: Spotify songs

These are the same songs as on the Week 8 page (lecturemenu8.html). In February 2023, Spotify and YouTube statistics were collected for the top 10 Spotify songs of each of 2,079 popular artists (the "Spotify and Youtube" dataset on Kaggle). A song credited to several artists appears once for each of them, so we count each song once, which leaves **18,860 different songs**. One row is one song, and our variable is

- X = the length of a song, in minutes.

The first five rows:

Song	Artist	Minutes
Feel Good Inc.	Gorillaz	3.71

Song	Artist	Minutes
Rhinestone Eyes	Gorillaz	3.34
New Gold	Gorillaz	3.59
On Melancholy Hill	Gorillaz	3.90
Clint Eastwood	Gorillaz	5.68
...	...	...

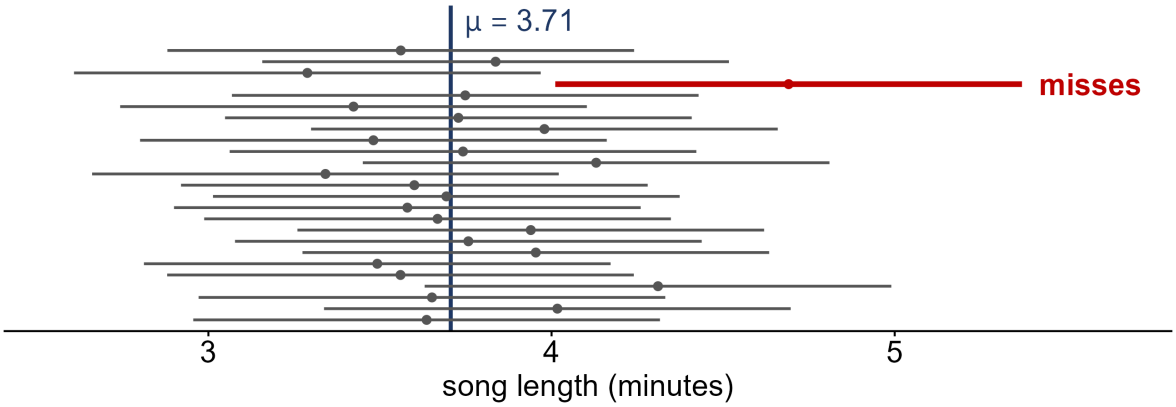
Usually we see only a sample, but here we have every song, so we can **treat the 18,860 songs as the population** and compute its parameters directly:

$$\mu = 3.71 \text{ minutes}, \qquad \sigma = 1.51 \text{ minutes}.$$

Song length is skewed right: most songs run 2.5 to 5 minutes, but 41 are longer than 10 minutes (the Week 8 page has the histogram). Knowing μ lets us check whether an interval captured it, which is never possible with real data.

Example: how often does an interval capture μ?

We can check the method here, because we know μ. Below are 25 random samples of $n = 20$ songs. For each one the standard error is $\sigma/\sqrt{n} = 1.51/\sqrt{20} \approx 0.34$ minutes, and each line is $\bar{x} \pm 2(0.34)$, the sample mean plus or minus the margin of error.



How many of the 25 intervals capture $\mu = 3.71$? Count the lines that cross the vertical line at 3.71: **24 of the 25** do, and one (in red) misses. That is 96%, close to the 95% we expect: about 95% of all samples put $\bar{x}$ within 2 standard errors of μ, and exactly those samples give an interval that reaches μ. Each new set of 25 samples would give a slightly different count, but about 95% in the long run.

Example: an interval for mean song length

In practice we see only one sample. Take the bottom interval in the plot, which came from these $n = 20$ songs, drawn at random from the 18,860:

Song	Artist	Minutes	Song	Artist	Minutes
Wish You Were Here	Pink Floyd	5.58	Tangled up in You	Staind	4.60
Told You So	Mike Perry	2.44	Tequila Sunrise	Cypress Hill	4.30
Family Tradition	Hank Williams, Jr.	4.05	Bad Memories	Elley Duhé	2.48
BUONANOTTE	Ernia	3.67	Stop! In The Name Of Love	The Supremes	2.88
LIKE ME	42 Dugg	2.22	WITHOUT YOU	The Kid LAROI	2.69
Ayer Me Llamó Mi Ex	KHEA	3.45	Paris Ka Trip	Yo Yo Honey Singh	3.27
B.I.B.L.E. (Basic Instructions Before Leaving Earth)	GZA	4.55	Perreo en La Luna	Dalex	3.64
Our Last Summer	Meryl Streep	2.96	Ai Preto	L7NNON	2.19
Roll Wit Me	Boyz II Men	3.53	Sumérgeme	Jesús Adrián Romero	5.46
When You Say Nothing At All	Ronan Keating	4.28	You Used to Love Me	Faith Evans	4.49

Their mean length is

$$\bar{x} = \frac{5.58 + 2.44 + \cdots + 4.49}{20} = \frac{72.73}{20} \approx \mathbf{3.64}\ \textbf{minutes},$$

and $\sigma = 1.51$ minutes is known from the population.

a. *What is the 95% confidence interval for μ?* The standard error is

$$\frac{\sigma}{\sqrt{n}} = \frac{1.51}{\sqrt{20}} = \frac{1.51}{4.47} \approx 0.34 \text{ minutes},$$

so the margin of error is $2(0.34) = 0.68$ and the interval is

$$3.64 \pm 0.68, \quad \text{from } \mathbf{2.96} \text{ to } \mathbf{4.32} \text{ minutes}$$

"We estimate with 95% confidence that the mean length of all 18,860 songs is between 2.96 and 4.32 minutes."

b. *Does it capture $\mu = 3.71$?* **Yes**: $2.96 \le 3.71 \le 4.32$.

c. *If we did not know μ, what would "95% confident" mean?* The 95% describes the **method**, not this one interval. About 95% of samples give an interval that captures μ. Once an interval is computed, it either contains μ or it does not. It is **not** correct to say there is a 95% probability that μ is between 2.96 and 4.32.

A confidence interval for a proportion

The interval for a mean was point estimate $\pm$ 2 standard errors. The same idea works for a proportion. By the CLT, for large n the sample proportion $\hat{p}$ is approximately normal with mean p and standard error $\sqrt{p(1-p)/n}$, so about 95% of samples give a $\hat{p}$ within 2 standard errors of p. Turning that around, as we did for the mean:

$$\hat{p} \ \pm \ 2\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

is a 95% confidence interval for p. There is one change from the formula for the standard error: we do not know p (that is what we are estimating), so the standard error uses $\hat{p}$ in its place.

When it works. The interval leans on the CLT, so check that there are enough successes **and** failures in the sample: $n\hat{p} > 5$ and $n(1-\hat{p}) > 5$.

The data: teens and cell phones

In Week 1 we met a Pew Research Center survey that asked 743 U.S. teens aged 13 to 17, "Are you losing focus in class by checking your cell phone?" The published counts are:

Response	Teens
Never	289
Rarely	216
Sometimes	178
Often	60
Total	743

Call a teen a "success" if they answered "sometimes" or "often":

- X = 1 if a teen loses focus in class sometimes or often, and 0 otherwise.

Example: losing focus in class

Estimate p, the proportion of **all** U.S. teens who lose focus in class sometimes or often.

- **Point estimate.** $178 + 60 = 238$ of the $n = 743$ teens, so $\hat{p} = 238/743 \approx 0.320$.
- **Check.** There are 238 successes and 505 failures, both far more than 5.
- **Standard error.** $\sqrt{0.320 \times 0.680/743} \approx 0.017$.
- **Interval.** $0.320 \pm 2(0.017) = 0.320 \pm 0.034$, from **0.286 to 0.354**.

We would report it as: **"We estimate with 95% confidence that between 28.6% and 35.4% of U.S. teens lose focus in class sometimes or often by checking their phones."** As with the mean, the 95% describes the method: about 95% of samples of 743 teens would give an interval that captures p.

Confidence levels and critical values

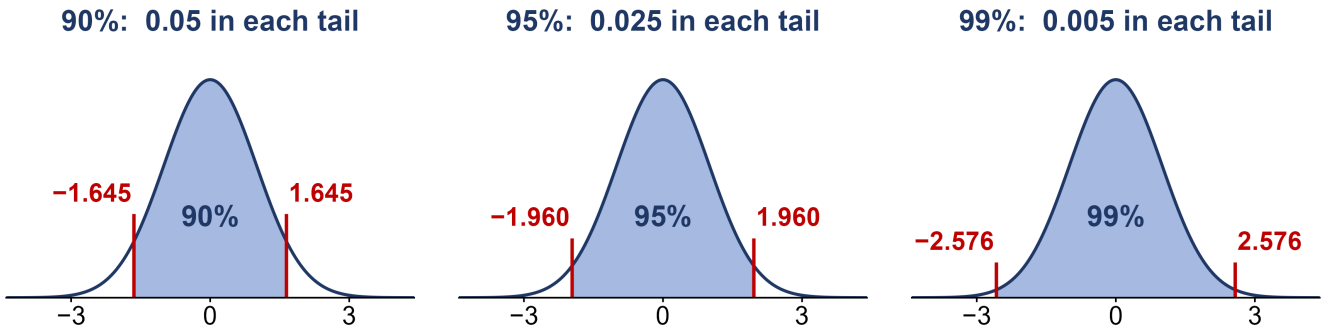
Every interval so far has been **estimate $\pm$ 2 standard errors**, which captures the parameter about 95% of the time. Suppose a cardiologist wants to be 99% sure instead. Should the interval get wider or narrower? Wider: to catch the parameter more often, we need a bigger net. The thing that changes is the 2, which becomes a larger number.

- **Confidence level** C. The proportion of all possible samples whose interval captures the parameter, such as 90%, 95% or 99%. It describes the method **before** the data are gathered.
- α. The proportion of samples whose interval misses: $\alpha = 1 - C$. It is split evenly between the two tails of the normal curve, $\alpha/2$ in each.

- **Critical value** $z_{1-\alpha/2}$. The z with area $1-\alpha/2$ to its left. A $100(1-\alpha)\%$ confidence interval is

$$\text{estimate} \;\pm\; z_{1-\alpha/2} \times SE$$

For a mean this is $\bar{x} \pm z_{1-\alpha/2}\,\sigma/\sqrt{n}$; for a proportion it is $\hat{p} \pm z_{1-\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$.



Why look up $1-\alpha/2$? The table gives the area to the **left** of z. To the left of the upper cut-off lies the middle C plus the lower tail $\alpha/2$, which is $1-\alpha/2$ in all.

Example: critical values

Find the critical value for 80%, 90% and 99% confidence. For each one, find α, then look for the area $1-\alpha/2$ **inside** the z-table and read off z from the row and column:

- **80%.** $\alpha = 0.20$, $\alpha/2 = 0.10$, so look for 0.90. The closest table value is 0.8997 in row 1.2, column .08: $z_{0.90} = \mathbf{1.28}$.
- **90%.** $\alpha = 0.10$, $\alpha/2 = 0.05$, so look for 0.95. It sits exactly halfway between 0.9495 ($z = 1.64$) and 0.9505 ($z = 1.65$), so $z_{0.95} = \mathbf{1.645}$.
- **95%.** $\alpha/2 = 0.025$, so look for 0.975: row 1.9, column .06 gives $z_{0.975} = \mathbf{1.96}$.
- **99%.** $\alpha = 0.01$, $\alpha/2 = 0.005$, so look for 0.995. It sits between 0.9949 ($z = 2.57$) and 0.9951 ($z = 2.58$); the more precise value is $z_{0.995} = \mathbf{2.576}$.

Confidence level	α	Area to the left, $1-\alpha/2$	Critical value $z_{1-\alpha/2}$
80%	0.20	0.90	1.28
90%	0.10	0.95	1.645
95%	0.05	0.975	1.960
99%	0.01	0.995	2.576

- **Try it out.** Give the confidence level and critical value for $\alpha = 0.25$, $\alpha = 0.08$ and $\alpha = 0.03$.
  - $\alpha = 0.25$: 75% confidence; look up $1 - 0.125 = 0.875$, which is closest to 0.8749 at $z = 1.15$.

- $\alpha = 0.08$: 92% confidence; look up $1 - 0.04 = 0.96$, which is closest to 0.9599 at $z = 1.75$.
- $\alpha = 0.03$: 97% confidence; look up $1 - 0.015 = 0.985$, which is 0.9850 at $z = 2.17$.

An interval for a proportion at other confidence levels

The data: a gene and a drug

Clopidogrel (brand name Plavix) is a blood thinner given to people after a heart attack or when a stent is put in a heart artery. It is a *prodrug*: it does nothing until the liver enzyme **CYP2C19** switches it on. The gene for that enzyme comes in several versions. People who carry the **CYP2C19*2** variant (a single DNA letter change, G to A, at the site rs4244285) make a version of the enzyme that does not work, so they switch on less of the drug. For heart patients who carry one or two copies, the current clinical guideline (CPIC, 2022) recommends a different drug.

The **1000 Genomes Project** sequenced the genomes of 2,504 people from 26 populations around the world to catalog human genetic variation. We use its 504 East Asian participants (Han Chinese in Beijing, Southern Han Chinese, Japanese in Tokyo, Kinh in Ho Chi Minh City, and Chinese Dai in Xishuangbanna). The first five:

Person	Population	Genotype	Copies of *2	X
HG00403	Southern Han Chinese	G/G	0	0
HG00404	Southern Han Chinese	A/A	2	1
HG00406	Southern Han Chinese	G/A	1	1
HG00407	Southern Han Chinese	G/G	0	0
HG00409	Southern Han Chinese	G/G	0	0
…	…	…	…	…

Each person has two copies of the gene, one from each parent, so the genotype is two letters: G/G has no copies of *2, G/A has one and A/A has two.

- X = 1 if a person carries at least one copy of *2, and 0 otherwise.

Example: carriers of CYP2C19*2

Of the $n = 504$ people, 239 carry one copy of *2 and 38 carry two, so 277 are carriers. Estimate p, the proportion of **all** East Asian people who carry the variant, with 90%, 95% and 99% confidence.

a. *Point estimate, check and standard error.*

- $\hat{p} = 277/504 \approx 0.550$.
- **Check.** There are 277 successes and 227 failures, both far more than 5.
- **Standard error.** $\sqrt{0.550 \times 0.450/504} = \sqrt{0.000491} \approx 0.022$.

b. *The three intervals.* Only the critical value changes:

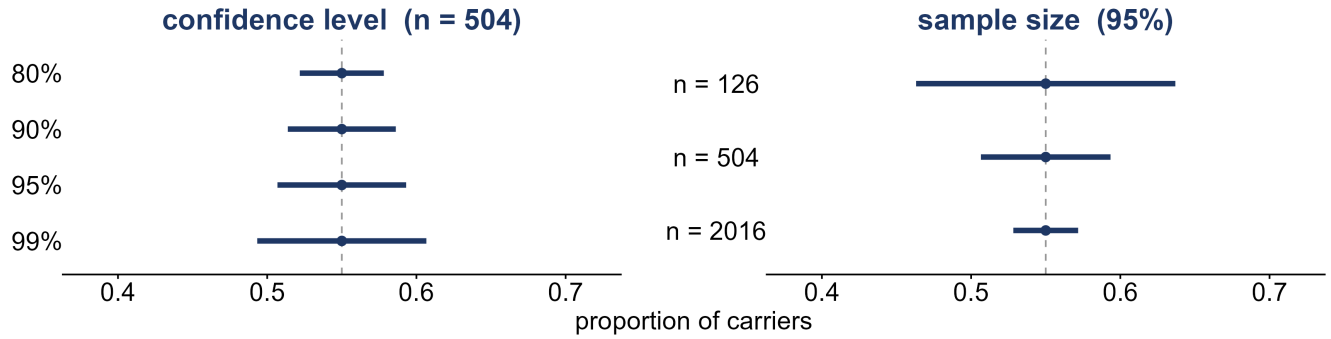
Confidence level	Critical value	Margin of error	Interval
90%	1.645	1.645(0.022) = 0.036	0.550 ± 0.036: **0.514 to 0.586**
95%	1.960	1.960(0.022) = 0.043	0.550 ± 0.043: **0.507 to 0.593**
99%	2.576	2.576(0.022) = 0.057	0.550 ± 0.057: **0.493 to 0.607**

For example, the 90% interval is reported as: **"We estimate with 90% confidence that between 51.4% and 58.6% of East Asian people carry at least one copy of CYP2C19*2."**

c. *Can we say that more than half of East Asian people carry *2?* At 90% and 95% confidence, yes: every value in those intervals is above 0.5. At 99% confidence, no: the interval reaches down to 0.493, so 0.5 is still a plausible value. Asking for more confidence costs precision.

Width, confidence level and sample size

The margin of error is $z_{1-\alpha/2} \times SE$, so two things control the width of an interval: the confidence level, through z, and the sample size, through the standard error. The plot shows the carrier interval at four confidence levels (left), and the 95% interval as it would be with 4 times fewer or 4 times more people at the same $\hat{p}$ (right).



- **Higher confidence, wider interval.** A bigger C means a bigger critical value.
- **Larger sample, narrower interval.** The standard error has $\sqrt{n}$ on the bottom, so 4 times the sample size gives **half** the width.

So there is a trade-off: for a fixed n, more confidence costs precision. The way to get both is more data.

Intervals for a mean when σ is unknown

Every interval for μ so far used σ, the **population** standard deviation:

$$\bar{x} \pm z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}$$

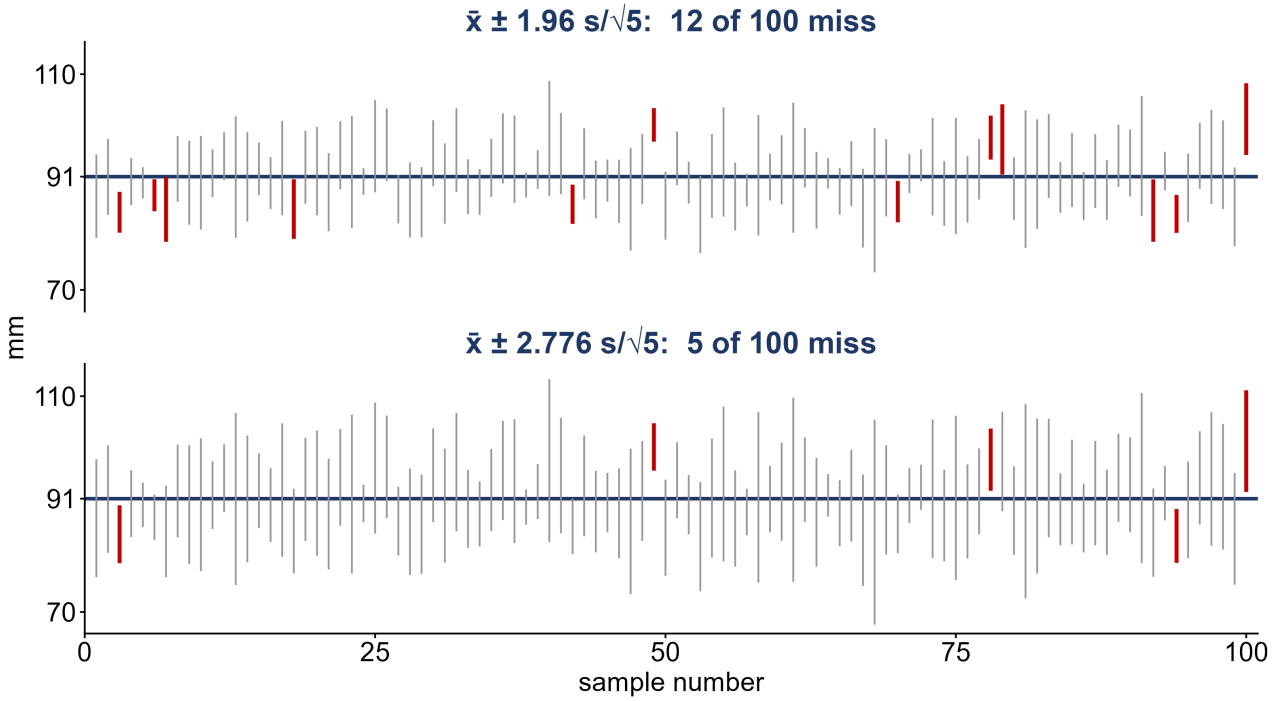
But if we do not know μ, we almost never know σ either. The natural fix is to put the sample standard deviation s in its place:

$$\bar{x} \pm z_{1-\alpha/2}\frac{s}{\sqrt{n}}$$

Unfortunately, this interval captures μ **less often** than the confidence level promises, especially when n is small. The reason is that s itself changes from sample to sample. When a sample happens to have a small s, its interval is too narrow, and those narrow intervals miss μ more often.

Does s in place of σ work?

Suppose seal whisker lengths are normal with $\mu = 91$ mm and $\sigma = 8$ mm (made-up values, close to the spotted seals below). We simulate 100 samples of only $n = 5$ whiskers each, and from each one build the interval $\bar{x} \pm 1.96\, s/\sqrt{5}$. A 95% method should miss about 5 times. The top panel shows the result; misses are in red.



With 1.96, **12 of the 100** intervals miss μ, more than twice the 5 we expect. (In the long run this method captures μ only about 88% of the time when $n = 5$.) The bottom panel uses the same 100 samples with a larger multiplier, 2.776, and **5 of the 100** miss, just as a 95% method should. That larger multiplier comes from the t distribution.

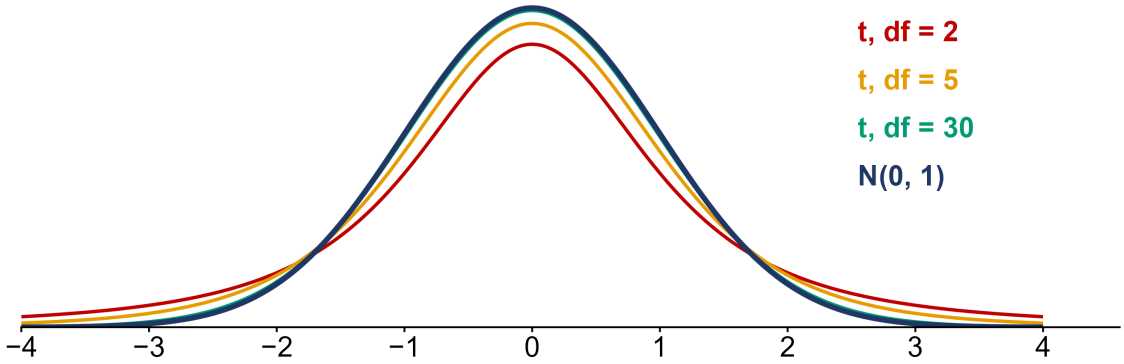
The t distribution

When we standardize $\bar{x}$ with s instead of σ, the result

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

does not follow the standard normal distribution. It follows a t **distribution**.

- t **distribution.** A bell-shaped curve centered at 0, like $N(0, 1)$, but with **heavier tails**: more of its area is far from 0. It accounts for the extra uncertainty from estimating σ with s.
- **Degrees of freedom** (df). The t distribution is a family of curves, one for each value of $df = n - 1$. The more degrees of freedom, the closer t is to $N(0, 1)$.
- t **critical value** $t_{n-1,\,1-\alpha/2}$. The value with area $1 - \alpha/2$ to its left under the t distribution with $n - 1$ degrees of freedom, read from the t-table.



As n grows, s becomes a more and more reliable estimate of σ, and the t distribution approaches the standard normal. By about $df = 30$ the two are nearly indistinguishable, and a t distribution with infinitely many degrees of freedom **is** the standard normal.

The interval for μ when σ is unknown is

$$\bar{x} \pm t_{n-1,\,1-\alpha/2}\,\frac{s}{\sqrt{n}}$$

Strictly, it assumes the population is normal, but it works well for most populations as long as they are not very skewed or full of outliers, and for any population once $n \geq 30$.

Reading the t-table

The course t-table (misc/t_table.html) is organized differently from the z-table. The **rows** are the degrees of freedom, the **columns** are confidence levels (with the matching upper-tail and two-tail areas in the header), and the values **inside** the table are the critical values. Here is part of it:

df	80%	90%	95%	99%
4	1.533	2.132	2.776	4.604
8	1.397	1.860	2.306	3.355
9	1.383	1.833	2.262	3.250
11	1.363	1.796	2.201	3.106
24	1.318	1.711	2.064	2.797
29	1.311	1.699	2.045	2.756
100	1.290	1.660	1.984	2.626

df	80%	90%	95%	99%
z	1.282	1.645	1.960	2.576

The last row is the standard normal: the critical values t is closing in on as df grows.

Example: reading the t-table

a. *95% confidence with* $n = 10$. $df = 10 - 1 = 9$; row 9, column 95% gives $t = \mathbf{2.262}$.

b. *90% confidence with* $n = 25$. $df = 24$; row 24, column 90% gives $t = \mathbf{1.711}$.

c. *99% confidence with* $n = 10$. $df = 9$; row 9, column 99% gives $t = \mathbf{3.250}$.

Compare these with z for the same confidence levels, 1.960, 1.645 and 2.576: the t value is **always larger**, so the interval is wider. That extra width is what restores the promised confidence level.

- **Try it out.** Use the full t-table (misc/t_table.html) to check these critical values.

Confidence level	Sample size	$df = n - 1$	α	Critical value $t_{n-1,\,1-\alpha/2}$
90%	$n = 3$	2	0.10	2.920
85%	$n = 5$	4	0.15	1.778
95%	$n = 10$	9	0.05	2.262
99%	$n = 14$	13	0.01	3.012
80%	$n = 22$	21	0.20	1.323

Which interval?

Estimating	Interval	When it works
a proportion p	$\hat{p} \pm z_{1-\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$	$n\hat{p} > 5$ and $n(1-\hat{p}) > 5$
a mean μ, σ known	$\bar{x} \pm z_{1-\alpha/2}\,\sigma/\sqrt{n}$	the population is normal, or $n \geq 30$
a mean μ, σ unknown	$\bar{x} \pm t_{n-1,\,1-\alpha/2}\,s/\sqrt{n}$	the population is roughly normal, or $n \geq 30$

The interval for a proportion always uses z: there is no separate σ to estimate, because the standard error comes from $\hat{p}$ itself.

Example: spotted seal whiskers

The data: seal whiskers

Seals and sea lions have the largest whiskers of any mammals, and they use them to track the water movements left by swimming fish. Ginter and colleagues (2012) measured the longest undamaged whisker of 92 seals and sea lions from 11 species, to study how whisker shape evolved. Only each species' sample size, mean and standard deviation were published:

Species	Seals (n)	Mean length (mm)	SD (mm)
Harp seal	16	80.6	10.8
Harbor seal	15	86.5	10.5
Ringed seal	16	79.6	10.9
Spotted seal	9	91.2	7.9
Gray seal	8	73.8	12.5
…	…	…	…

- X = the total length of a seal's longest whisker, in mm (one whisker per seal).

Working the example

Estimate μ, the mean whisker length of **all** spotted seals, with 95% confidence, from the 9 that were measured: $n = 9$, $\bar{x} = 91.2$ mm, $s = 7.9$ mm.

a. *z or t? How many degrees of freedom?* We know only s, not σ, so use t, with $df = 9 - 1 = 8$. Row 8, column 95% of the t-table gives $t = 2.306$.

b. *What is the standard error?*

$$\frac{s}{\sqrt{n}} = \frac{7.9}{\sqrt{9}} = \frac{7.9}{3} \approx 2.63 \text{ mm}$$

c. *What is the 95% confidence interval for μ?*

$$91.2 \pm 2.306(2.63) = 91.2 \pm 6.06, \quad \text{from } \mathbf{85.1} \text{ to } \mathbf{97.3} \text{ mm}$$

"We estimate with 95% confidence that the mean length of a spotted seal's longest whisker is between 85.1 and 97.3 mm." With $z = 1.96$ instead, the margin of error would have been only $1.96(2.63) = 5.15$ mm, too narrow for a sample of 9.

Example: watermelon citrulline

The data: watermelon citrulline

Korea's national gene bank stores seed of watermelon varieties (*Citrullus lanatus*) collected from around the world, so that plant breeders can use their genes. Assefa and colleagues (2020) grew 100 of these gene-bank varieties and measured each one's fruit for **citrulline**, an amino acid named after watermelon (*Citrullus*), which was first found in its juice. Citrulline is sold as a sports supplement, so breeders are interested in varieties that have a lot of it.

We take a random sample of 12 of the 100 varieties. The first five are:

Variety	Origin	Citrulline (mg/g)
IT120008	Taiwan	27.16
IT199814	Kazakhstan	25.83
IT203017	Russia	26.38
IT203029	Kazakhstan	24.01
IT203627	unknown	32.79
…	…	…

- X = the citrulline in a variety's fruit flesh, in mg per gram of dry weight.

Working the example

Estimate μ, the mean citrulline content of all 100 gene-bank varieties, with 90% and with 99% confidence.

First, the sample mean and standard deviation of the 12 values. The table lists each value x, its deviation from the mean $x - \bar{x}$, and the squared deviation:

Variety	x	$x - \bar{x}$	$(x - \bar{x})^2$
IT120008	27.16	0.09	0.01
IT199814	25.83	-1.24	1.53
IT203017	26.38	-0.69	0.47
IT203029	24.01	-3.06	9.35
IT203627	32.79	5.72	32.74
IT204167	25.98	-1.09	1.18
IT216860	21.20	-5.87	34.44
IT321075	24.79	-2.28	5.19
K192386	38.42	11.35	128.86
K192397	33.05	5.98	35.78
K192432	19.80	-7.27	52.83
K192469	25.41	-1.66	2.75
Sum	324.82	0.00	305.14

$$\bar{x} = \frac{324.82}{12} \approx 27.07 \text{ mg/g}, \qquad s = \sqrt{\frac{305.14}{12 - 1}} \approx 5.27 \text{ mg/g}$$

a. *Degrees of freedom and standard error.* σ is unknown, so use t with $df = 12 - 1 = 11$.

$$SE = \frac{s}{\sqrt{n}} = \frac{5.27}{\sqrt{12}} = \frac{5.27}{3.46} \approx 1.52 \text{ mg/g}$$

b. *The 90% and 99% intervals.* Row 11 of the t-table gives $t = 1.796$ for 90% and $t = 3.106$ for 99%:

- **90%:** $27.07 \pm 1.796(1.52) = 27.07 \pm 2.73$, from **24.34 to 29.80 mg/g**.
- **99%:** $27.07 \pm 3.106(1.52) = 27.07 \pm 4.72$, from **22.35 to 31.79 mg/g**.

"We estimate with 90% confidence that the mean citrulline content of the gene bank's 100 watermelon varieties is between 24.34 and 29.80 mg per gram of dry weight."

c. *Which is wider, and why?* The 99% interval: more confidence needs a larger critical value, so a larger margin of error from the same sample.

Here we can check our work, since all 100 varieties were measured: their mean is 27.65 mg/g, which both intervals capture.

Sources

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